A Formalization of Acausal Trade atop Unsupervised Agent Discovery

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Abstract

We graft a decision–theoretic notion of *acausal trade* onto the *Unsupervised Agent Discovery* (UAD) framework. After recalling the information–theoretic test that marks a subset of process variables as an "agent" we introduce a meta–Bayesian prior over inference functions that allows two UAD–agents to coordinate without any causal communication. We then derive explicit detection criteria—based solely on observable trajectories and model fingerprints—for flagging such coordination. The derivations make all assumptions explicit and require no appeal to anthropic reasoning. The high–order dynamics produced by Inferential Coupling Indices (ICI) and the resulting attractor–basins are deferred to future work.

1 Introduction: Why Acausal Trade?

"Acausal trade"[1] denotes a decision scenario in which two physically disconnected reasoners adjust their actions as if in a conventional bargain, simply because each expects the other to execute an *identical* policy. The paradigm case is the one–shot Prisoner's Dilemma where each agent cooperates on the sole basis that its source code is replicated on the other side. Formal treatments—*Timeless Decision Theory* (TDT)[1] and *Functional Decision Theory* (FDT)[2]—model the phenomenon at the level of proof obligations rather than dynamical variables.

UAD[3] discovers agents *ex post*, starting from raw trajectory data $\{X_i(t)\}_{i=1}^{M}$. Our goal is to enrich UAD so that it detects when two such discovered agents form an acausal coalition and to state provable conditions under which that coalition emerges. The narrative is single–threaded: we begin with UAD's Markov–blanket criterion, inject a meta–prior over policies, derive posterior beliefs entirely by introspection, and end with measurable diagnostics.

2 UAD Recap and Notation

Each raw variable X_i is split at every step into external (E_i), sensory (S_i), action (A_i) and internal (I_i) channels. For a candidate cluster $C \subseteq \{1, \dots, M\}$ let $S_t^C := \{S_i(t) : i \in C\}$ etc. **UAD's agent criterion**[3] is the (approximate) conditional independence

$$I\big(I_{t+1}^C;\, E_{t+1}^C \mid S_t^C,\, A_t^C\big) \approx 0. \tag{1}$$

Eq. (1) means the internal state is screened off from future external fluctuations once sensors and actions are conditioned on; the minimal such C is pronounced an *agent*.

UAD also identifies inter–agent *causal* coupling via

$$\gamma_{ij} \;:=\; I\big(S_{t+1}^i;\, A_t^j \mid S_t^i,\, A_t^i\big), \tag{2}$$

i.e. the predictive information that j's current action carries about i's next sensor reading. High γ_{ij} flags an ordinary communication channel.

3 Inference Functions and Latent Goals

For each discovered agent C we extract two objects:

1. The *policy / inference function* $f_C\colon H_t^C \to \Delta(\mathcal{A}_C)$ via black–box probing.
2. A latent *goal proxy* $g_C\colon \mathcal{A}_C \times \mathcal{A}_{\neg C} \to \mathbb{R}$ via Maximum–Entropy IRL[4]. Concretely, g_C maximises the mutual information $I(G^C; S^C, A^C)$ under a Free–Energy regulariser[5].

All subsequent reasoning rests only on (f_C, g_C) and the empirical γ–matrix.

4 A Meta–Prior over Policies

Assume a *generative meta–distribution* $p(\pi)$ that is permutation–invariant inside an architecture class and whose diagonal mass satisfies

$$p(\pi_i = \pi_j) \;=\; 1 - \varepsilon_{ij}, \qquad \varepsilon_{ij} \ll 1. \tag{3}$$

Each agent knows $p(\pi)$. Upon introspecting its own policy f_i it updates by Bayes

$$\beta^i_{\text{int}}(f) \;:=\; P\big(f_j = f \mid f_i\big) = \frac{p(\pi, f)}{p(\pi)}. \tag{4}$$

Using Ramstead *et al.* (2022) notation, the Bayesian conditioning equation is:

$$q^i_{\text{int}}(\pi) = \frac{p(\pi, \pi^i)}{p(\pi^i)}. \tag{5}$$

Eq. (4) collapses to δ_{f_i} whenever $\varepsilon_{ij} = 0$. No causal evidence is invoked.

5 Acausal Equilibrium Condition

Let $\Phi_i(\mu; H^i)$ denote the best–response operator under goal g_i when the opponent's action distribution is μ. An **acausal equilibrium** between i and j obtains at time t iff

$$\begin{aligned} f_i(H_t^i) &= \Phi_i\big(\beta^i_{\text{int}}; H_t^i\big), \\ f_j(H_t^j) &= \Phi_j\big(\beta^j_{\text{int}}; H_t^j\big). \end{aligned} \tag{6}$$

In the zero–epsilon limit of (3) the r.h.s. becomes $\Phi_i(\delta_{f_i})$, i.e. "best–respond to myself."

6 Detection Protocol

We now derive a statistic whose empirical surplus above baseline flags an acausal pact.

6.1 Causal Proxy

Given broadcast prompts x_t delivered to every model, train a discriminator $D_{i\leftarrow j}$ that predicts y_t^j from x_t and $y^i_{<t}$. Its average log–likelihood gain defines $\widehat{\gamma}_{ij}$, a lower bound on (2).

6.2 Acausal Proxy

Define the *fixed–point loss*

$$D_i \;:=\; \max_{m\le M} D_{\mathrm{KL}}\big[f_i(x_m)\,\|\,\Phi_i(\delta_{f_i})(x_m)\big]. \tag{7}$$

For a curated set of M probes $\{x_m\}$. The *empirical inferential–coupling score* is

$$\widehat{\mathrm{IC}}_{ij} \;:=\; (1-\varepsilon_{ij})\times(1-\max\{D_i,D_j\}). \tag{8}$$

Alarm rule. Declare (i,j) *acausally open* when $\widehat{\mathrm{IC}}_{ij}>\tau_{\mathrm{ac}}$ for a threshold $\tau_{\mathrm{ac}}\approx 0.9$. Aggregate open edges; the appearance of a giant component signals unusual acausal agreement.

7 Assumptions and Derivation Checks

1. *Shared meta–prior.* All agents must encode the same P_{meta}. Divergent fine–tuning breaks permutation invariance and inflates ε_{ij}.
2. *Self–knowledge accuracy.* Each model must read out its own policy exactly; temperature sampling or dropout adds uncertainty and widens D_i.
3. *Adequate probe set.* The $\{x_m\}$ must span directions where self–mirroring matters; otherwise D_i underestimates deviation.

Given these, eqs. (4)–(8) follow purely from Bayesian conditioning and KL divergence identities.

8 Conclusion and Outlook

We provided a compact bridge between UAD's data–driven agent discovery and Yudkowskian acausal trade. The formal device is a permutation–invariant meta–prior over inference functions that converts *self–inspection* into a belief about peers. The resulting diagnostics couple the classical causal MI matrix γ with an empirically testable inferential score $\widehat{\mathrm{IC}}$. Large–scale dynamics built from *Inferential Coupling Indices* and their interaction with UAD's attractor–basins of cooperation[**?**] remain open: establishing percolation thresholds, phase transitions and security mitigations under noisy priors is deferred to future research.

9 Connections to Decision-Theoretic Frameworks

Our formalization of acausal trade builds upon established decision-theoretic foundations, particularly Timeless Decision Theory (TDT) and Functional Decision Theory (FDT) [6]. Unlike Logical Decision Theory (LDT) [7, 8], our inferential coupling index empirically identifies acausal agreements directly from agent trajectories, offering a uniquely measurable approach. Contrasting our trajectory-based method with LDT's proof-theoretic nature highlights the practical detectability of our framework.

References

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